

CONTENT

CONTENT	1
DESCRIPTION	2
📌 Goal	2
🧩 Services Overview	2
① Custom (Business) Services	2
② Architecture Services	2
③ Infrastructure Services	3
④ External Services	3
💻 Available Consoles	3
🏗️ Architecture Summary	4
🎯 Project Benefits	4
DIAGRAM	5
EXAMPLE POSTMAN	6
EXAMPLE SWAGGER 3	22
EXAMPLE GRAFANA	34
EXAMPLE KAFKA-UI	42
CONFIGURATION OF EXTERNAL SERVICES	43
CONFIGURATION OF ENVIRONMENT VARIABLES	44

DESCRIPTION

Goal

The goal of this project is to demonstrate how to create an **API** for an **e-commerce** shop that sells **virtual products** (such as eBooks) using a **microservices architecture**.

The implementation is based on:

- **Java**
- **Spring Boot 3**
- **Spring Cloud**

The project showcases a complete microservices ecosystem including business services, infrastructure components, observability tools, and external integrations.

Services Overview

1 Custom (Business) Services

These services implement the core business logic of the e-commerce platform:

- **product-service**
Manages virtual products (e.g., eBooks).
 - **order-service**
Handles customer orders.
 - **payment-service**
Processes payments.
 - **delivery-service**
Manages delivery of virtual products.
 - **notification-service**
Sends email notifications to customers.
-

2 Architecture Services

These services support the microservices ecosystem and enable service coordination:

- **config-server**
Centralized configuration management for all services.
- **discovery-service**
Service registry that maintains a list of all running services in the ecosystem.

- **api-gateway**
Single entry point from the Internet to the microservices ecosystem.
Handles routing, filtering, and security at the edge.
-

3 Infrastructure Services

These services provide security, messaging, storage, and observability:

- **keycloak**
Manages authentication and authorization.
 - **kafka**
Enables asynchronous communication between services.
 - **kafka-ui**
Provides a user interface for monitoring Kafka topics and messages.
 - **postgres**
SQL database used for storing order data.
 - **mongo**
NoSQL database used for storing product data.
 - **grafana**
Dashboard UI for distributed logging, monitoring, and performance analysis.
 - **prometheus**
Collects and measures performance metrics.
 - **loki**
Aggregates logs from all services.
 - **tempo**
Collects and tracks distributed traces.
-

4 External Services

These third-party services are integrated into the system:

- **stripe**
External payment processing platform.
 - **mailtrap**
Email testing and delivery platform.
-



Available Consoles

The **ecommerce-virtual** project provides the following user interfaces:

Service	Purpose	URL
---------	---------	-----

Grafana (with Loki, Tempo, and Prometheus)	Distributed logging, monitoring & performance	http://localhost:3000
Kafka UI	Asynchronous communication monitoring	http://localhost:9000
Keycloak	Authentication & security management	http://localhost:7070
Stripe	Payment management	https://stripe.com
Mailtrap	Email testing	https://mailtrap.io

Architecture Summary

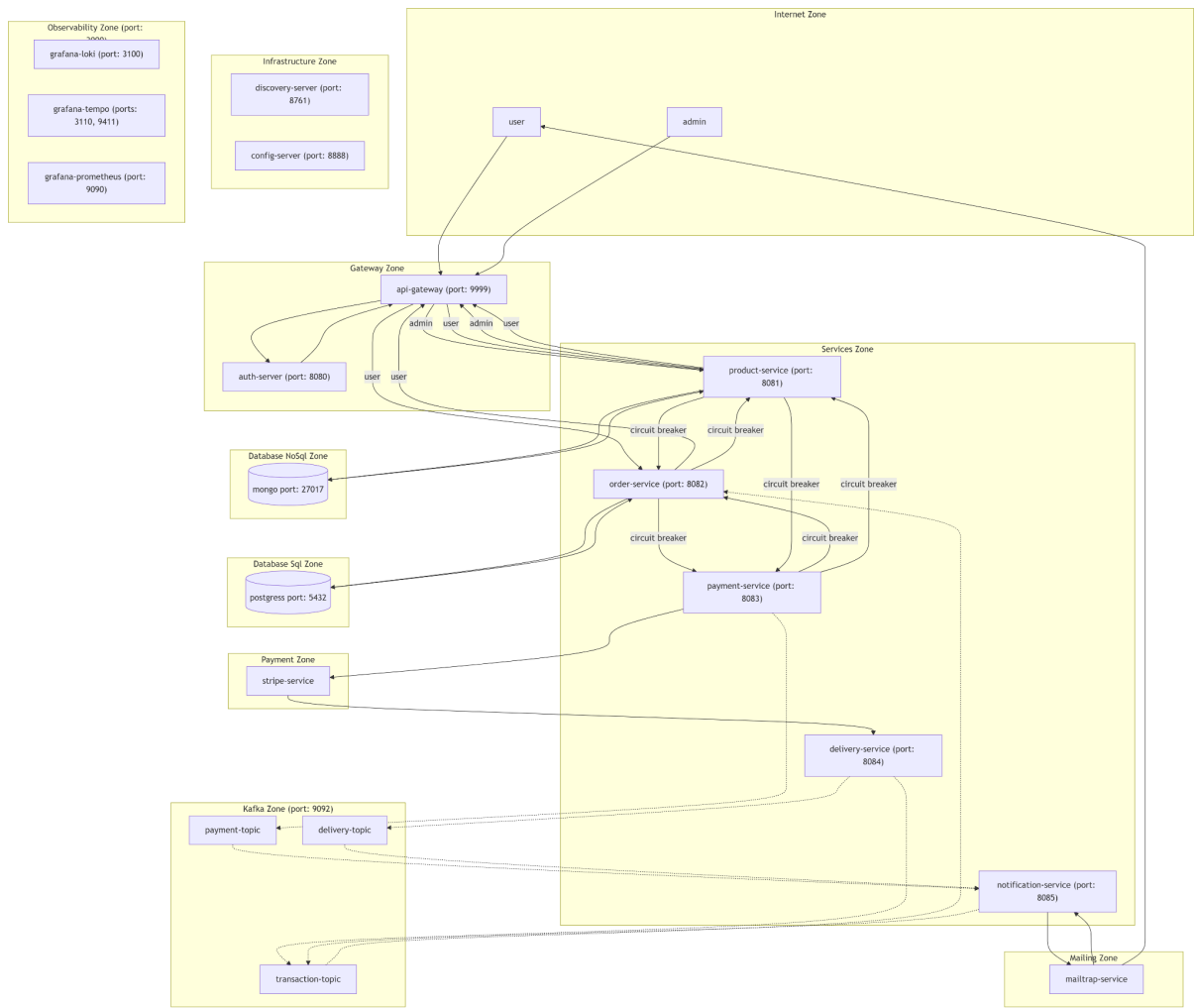
This project demonstrates:

- Microservices architecture
 - API Gateway pattern
 - Centralized configuration management
 - Service discovery
 - Event-driven communication using Kafka
 - Secure authentication and authorization with Keycloak
 - Observability using Prometheus, Grafana, Loki, and Tempo
 - Integration with external services (Stripe, Mailtrap)
 - Polyglot persistence (PostgreSQL + MongoDB)
-

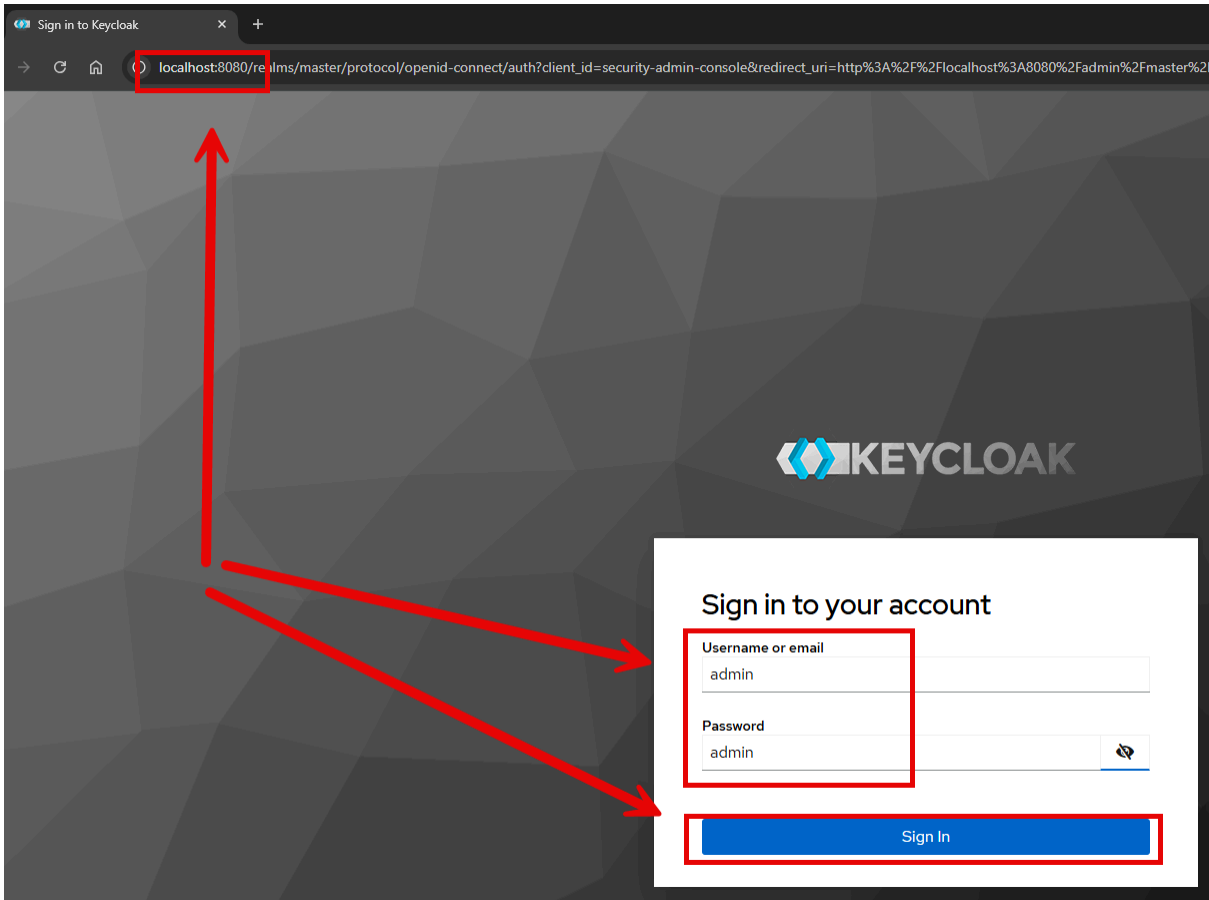
Project Benefits

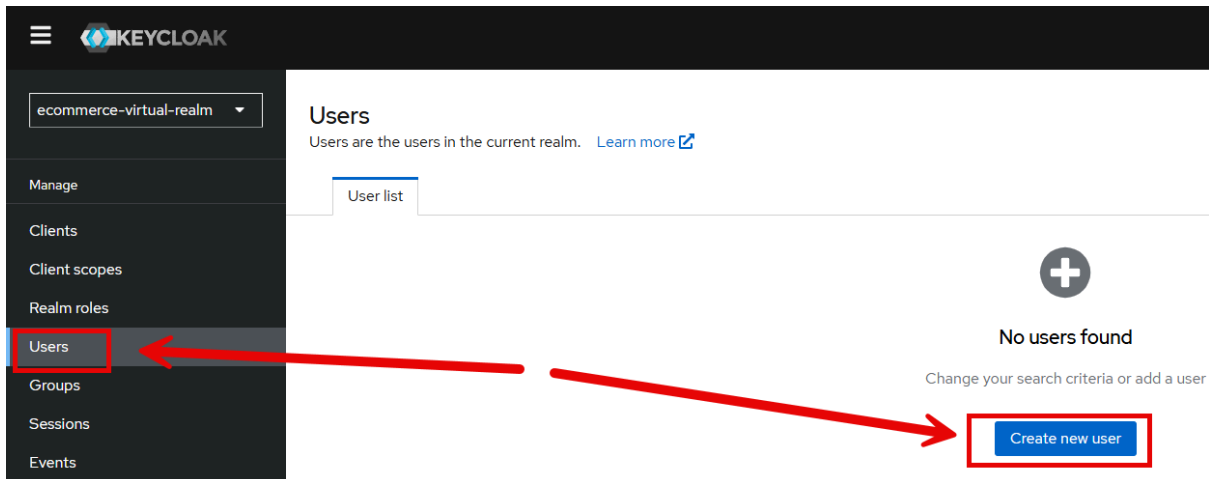
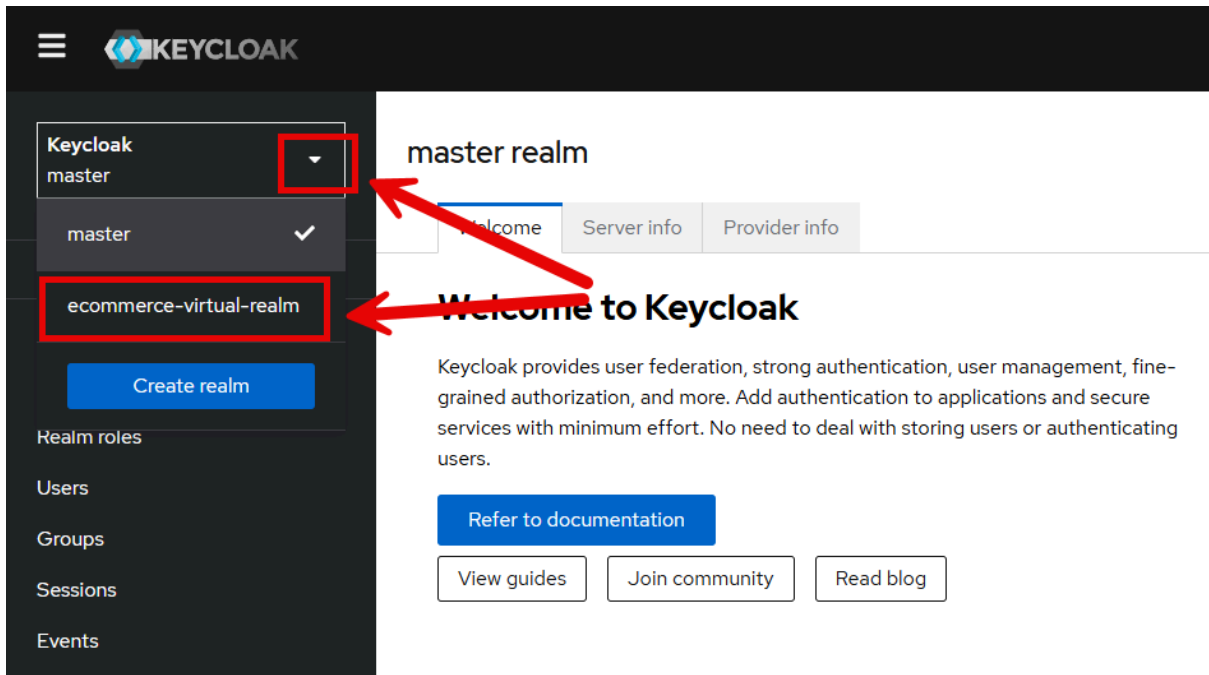
- Real-world microservices architecture example
- Full observability stack
- Secure authentication & authorization
- Event-driven communication
- Cloud-native ready design
- Clear separation of concerns between services

DIAGRAM



EXAMPLE POSTMAN





KEYCLOAK

ecommerce-virtual-realm

Manage

Clients

Client scopes

Realm roles

Users

Groups

Sessions

Events

Configure

Realm settings

Authentication

Identity providers

User federation

Users > Create user

Create user

Required user actions

Select action

Email verified

Off

General

Username

admin

Email

admin@example.com

First name

Admin

Last name

Admin

Groups

Join Groups

Jump to section

General

Create

Cancel

KEYCLOAK

ecommerce-virtual-realm

Manage

Clients

Client scopes

Realm roles

Users

Groups

Sessions

Events

Users > User details

Admin

Details

Credentials

Role mapping

Groups

Consents

Identity provider links

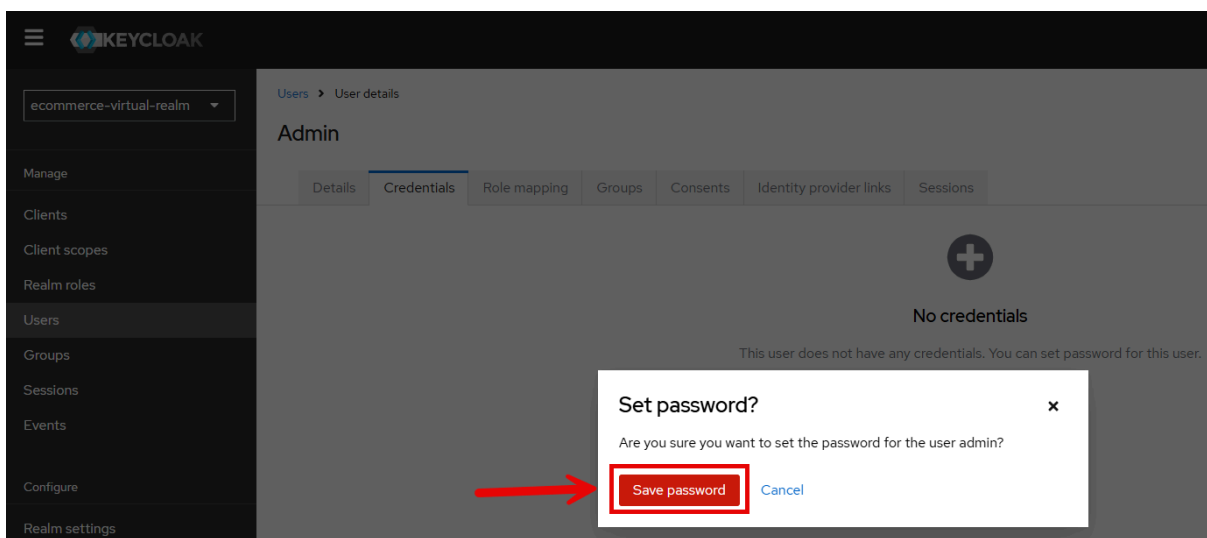
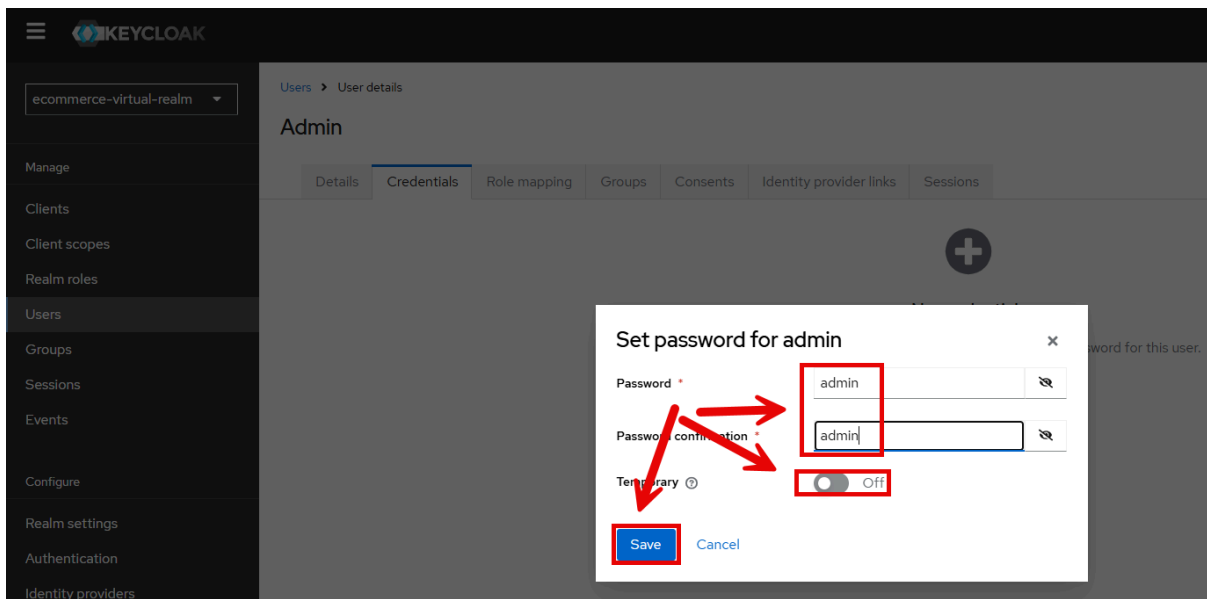
Sessions

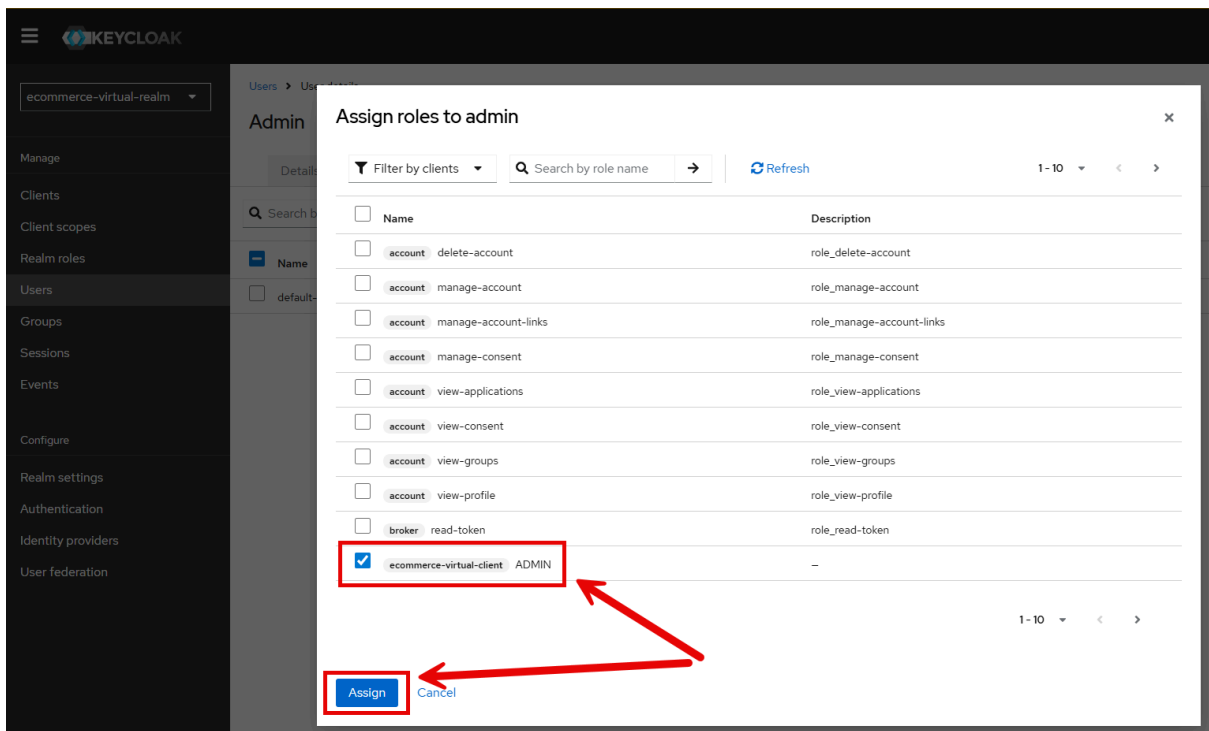
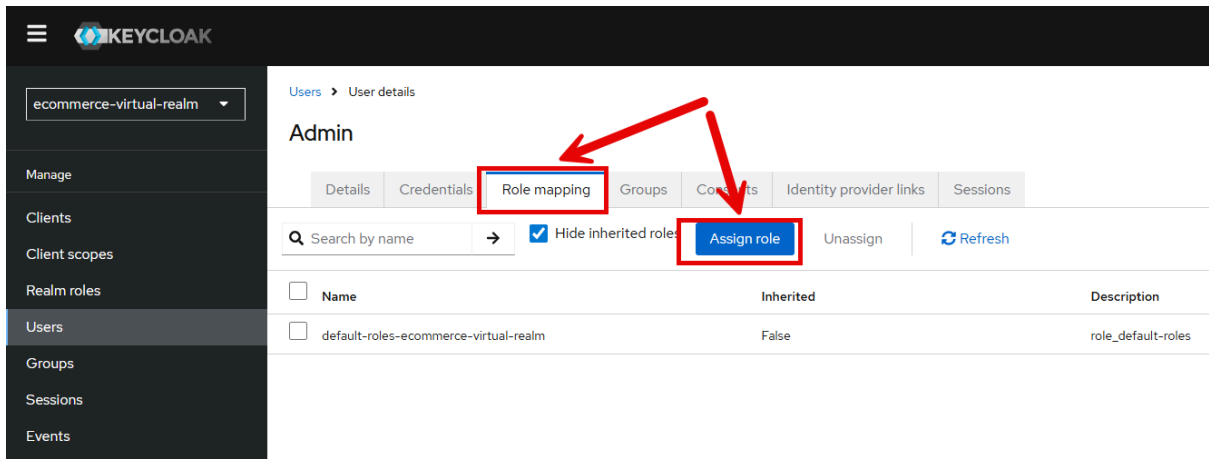
No credentials

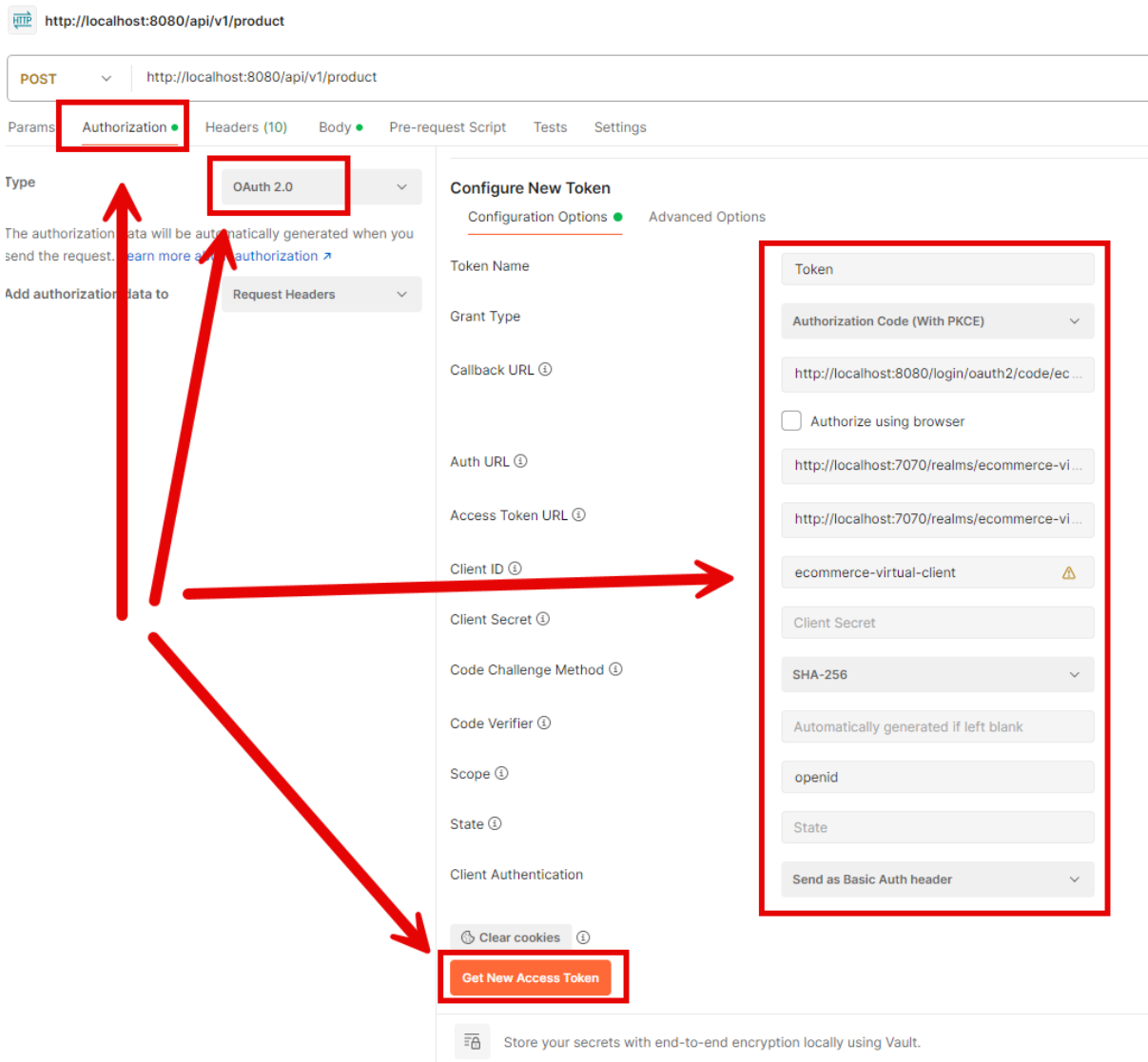
This user does not have any credentials. You can set password for this user.

Set password

Credential Reset







ECOMMERCE-VIRTUAL-REALM

Sign in to your account

Username or email

admin

Password

admin



Sign In

New user? [Register](#)

http://localhost:9999/api/v1/order

POST Save

Params Authorization Headers (10) Body Pre-request Script Tests Settings

none form-data x-www-form-urlencoded raw binary JSON

Body

raw

JSON

Send

cookies

Beautify

products: 1

http://localhost:9999/api/v1/order

POST Send

Params Authorization Headers (10) Body Pre-request Script Tests Settings

Type

The authorization code will be automatically generated when you send the request. [Learn more about authorization](#)

Add authorization data to Request Headers

OAuth 2.0

Configure New Token

Configuration Options Advanced Options

Token Name

Grant Type

Authorization Code (With PKCE)

Callback URL

http://localhost:9999/login/oauth2/code/ec...

☐ Authorize using browser

Auth URL

http://keycloak:8080/realms/ecommmerce-vi...

Access Token URL

http://keycloak:8080/realms/ecommmerce-vi...

Client ID

ecommerce-virtual-client

Client Secret

Client Secret

Code Challenge Method

SHA-256

Code Verifier

Automatically generated if left blank

Scope

openid

State

State

Client Authentication

Send as Basic Auth header

[Clear cookies](#)

[Get New Access Token](#)

ECOMMERCE-VIRTUAL-REALM

Sign in to your account

Username or email

Password



Sign In

New user? [Register](#)



ECOMMERCE-VIRTUAL-REALM

Register

* Required fields

Username *

Password *



Confirm password *



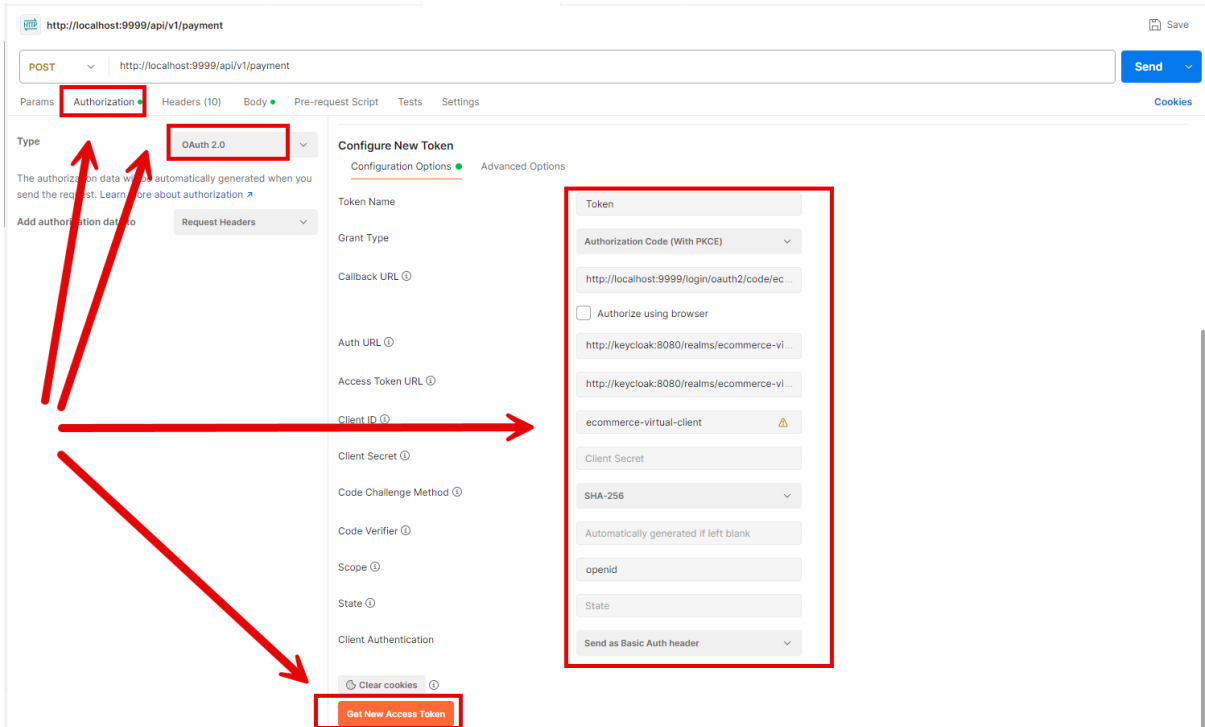
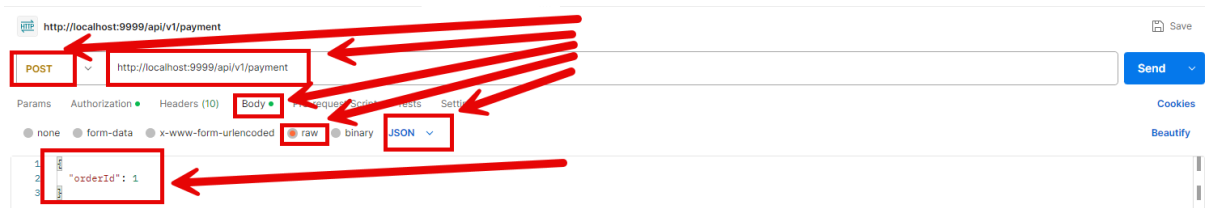
Email *

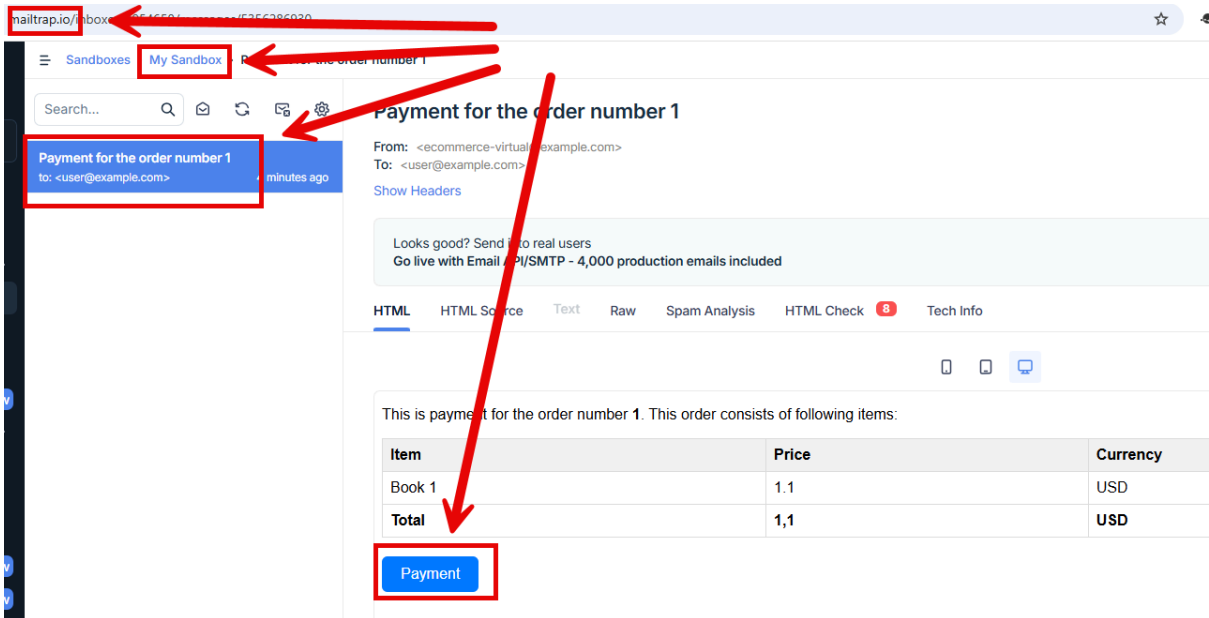
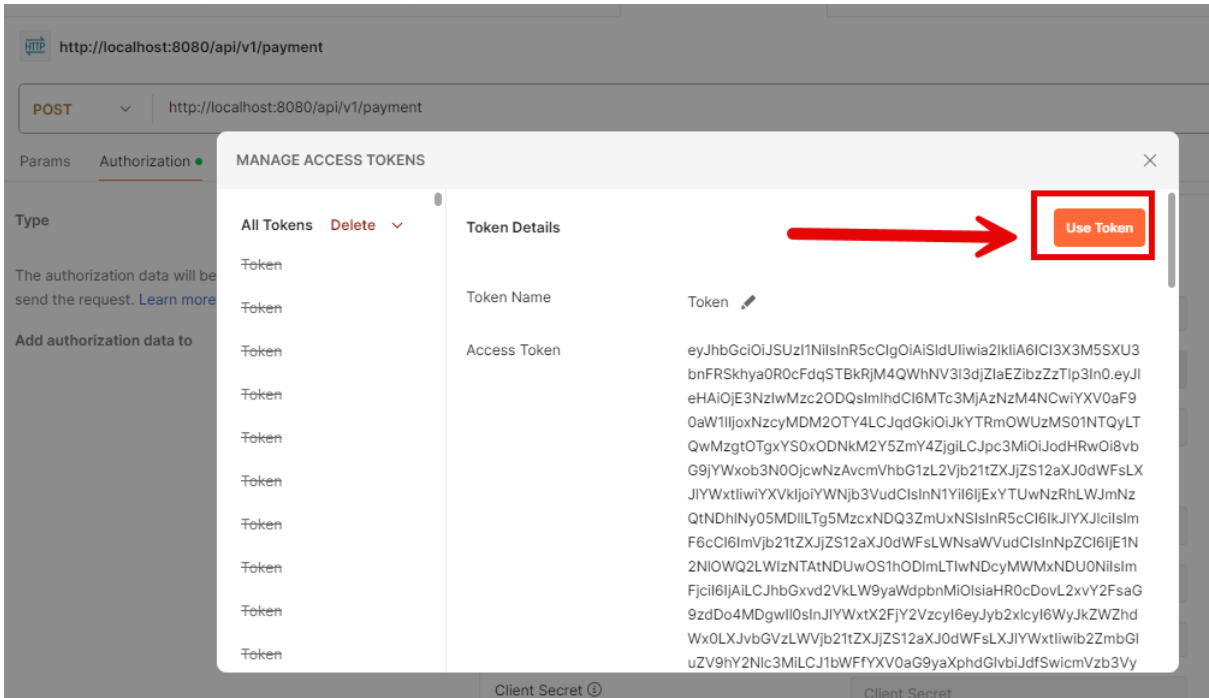
First name *

Last name *

Register

[« Back to Login](#)





Book 1

\$1.10

Pay with  link

OR

Contact information

Email

user@example.com

Payment method

 Card

Card information

4242 4242 4242 4242

VISA

12 / 34

567

123

Cardholder name

Krzysztof Wisniewski

Country or region

Poland

☐ Save my information for faster checkout

Pay securely at ecommerce-virtual-sandbox and
everywhere  link is accepted.

Pay

Powered by  | [Terms](#) | [Privacy](#)

← → ↻ 🏠 ⓘ localhost:8080/api/v1/delivery?orderId=1&token=5141259712

Pretty-print ☐

```
{"message": "Delivery successful for order 1"}
```

mailtrap.io/inboxes/...

Sandboxes My Sandbox

Search...

Delivery for the order number 1
to: <user@example.com> a minute ago

From: <ecommerce-virtual@example.com>
To: <user@example.com>

Show Headers

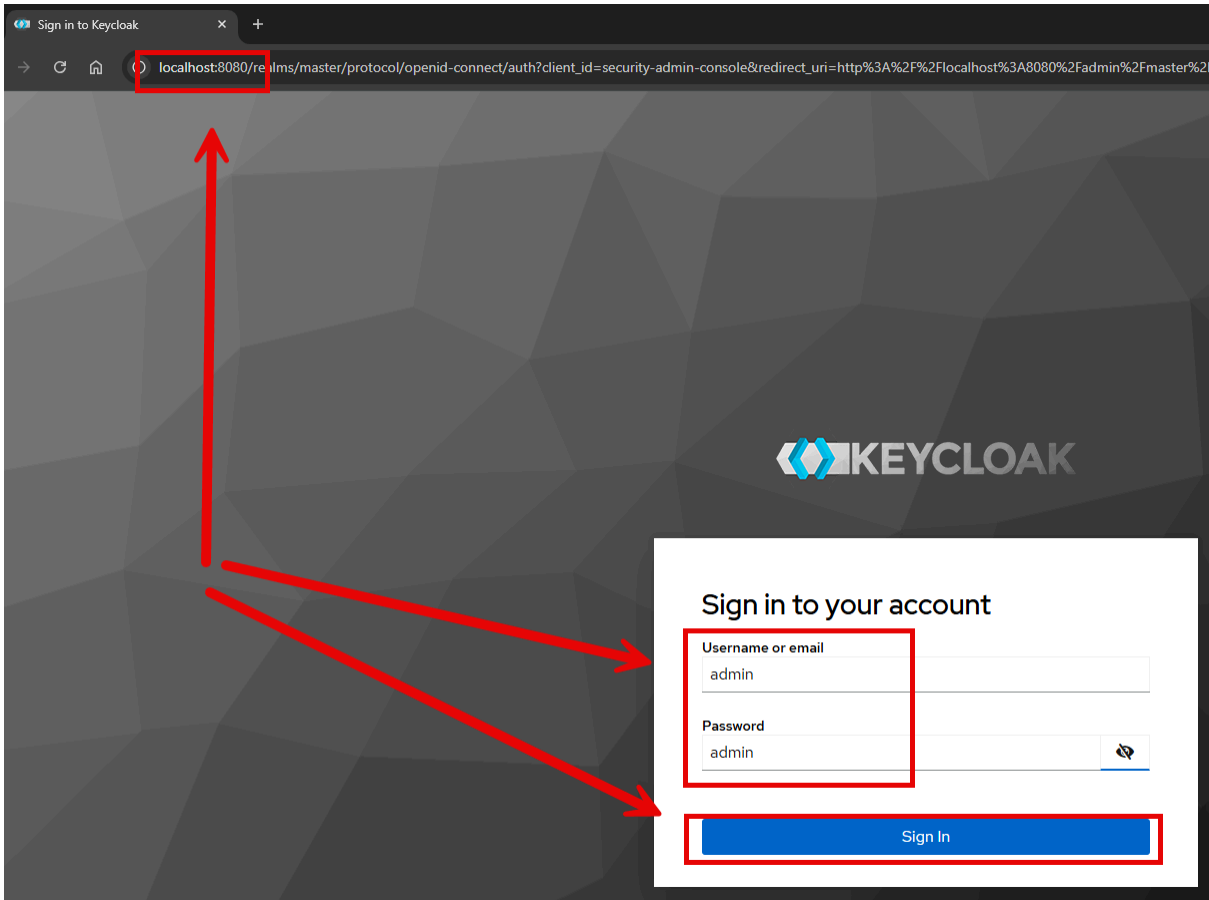
Looks good? Send it to real users
Go live with Email API/SMTP - 4,000 production emails included

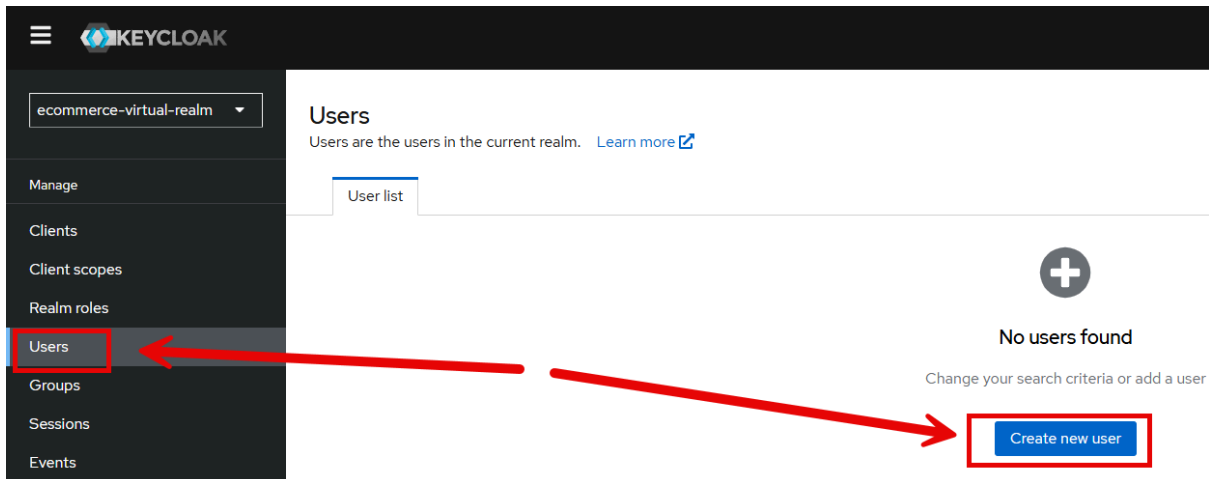
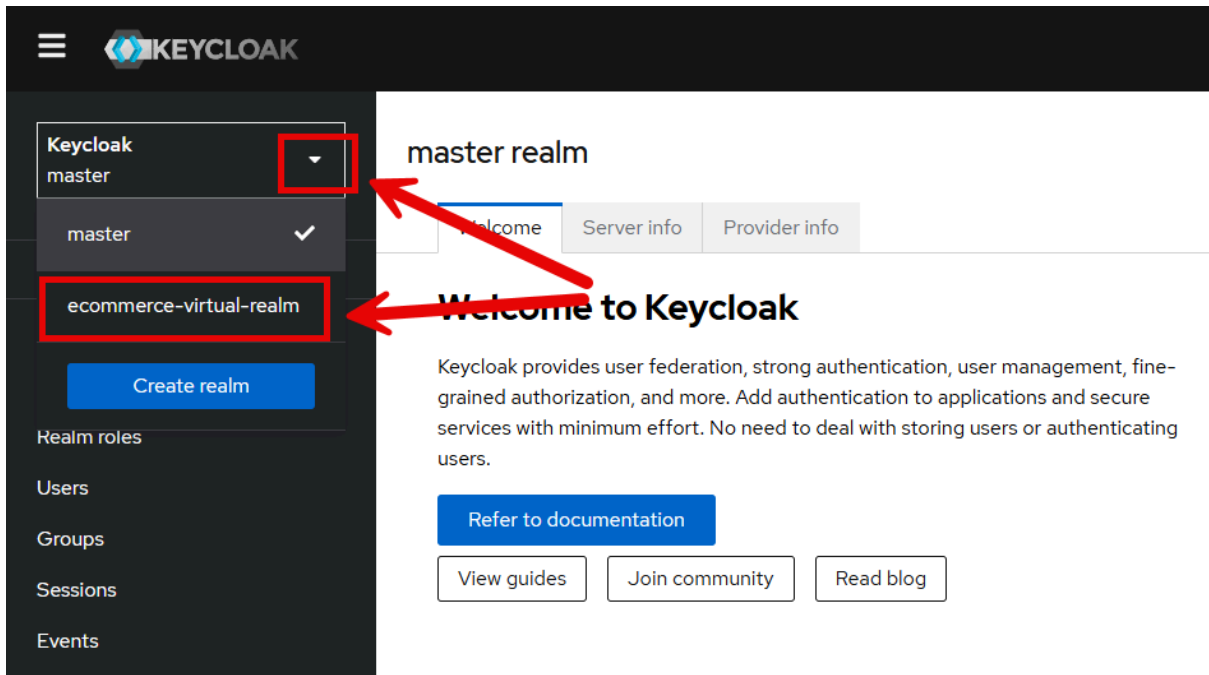
HTML HTML Source Text Raw Spam Analysis HTML Check 8 Tech Info

This is delivery for the order number 1. This order consists of following items:

Item	Download
Book 1	Download

EXAMPLE SWAGGER 3





KEYCLOAK

ecommerce-virtual-realm

Manage

Clients

Client scopes

Realm roles

Users

Groups

Sessions

Events

Configure

Realm settings

Authentication

Identity providers

User federation

Users > Create user

Create user

Required user actions ? Select action × ▼

Email verified ? ☐ Off

General

Jump to section

General

Username * admin

Email admin@example.com

First name Admin

Last name Admin

Groups ? [Join Groups](#)

Create

Cancel

KEYCLOAK

ecommerce-virtual-realm

Manage

Clients

Client scopes

Realm roles

Users

Groups

Sessions

Events

Users > User details

Admin

Details

Credentials

Role mapping

Groups

Consents

Identity provider links

Sessions

+

No credentials

This user does not have any credentials. You can set password for this user.

Set password

Credential Reset

KEYCLOAK

ecommerce-virtual-realm

Manage

Clients

Client scopes

Realm roles

Users

Groups

Sessions

Events

Configure

Realm settings

Authentication

Identity providers

Users > User details

Admin

Details

Credentials

Role mapping

Groups

Consents

Identity provider links

Sessions

+

Set password for admin

Close

Password *
admin

Password confirmation *
admin

Temporary ⓘ
☐ Off

Save Cancel

KEYCLOAK

ecommerce-virtual-realm

Manage

Clients

Client scopes

Realm roles

Users

Groups

Sessions

Events

Configure

Realm settings

Users > User details

Admin

Details

Credentials

Role mapping

Groups

Consents

Identity provider links

Sessions

+

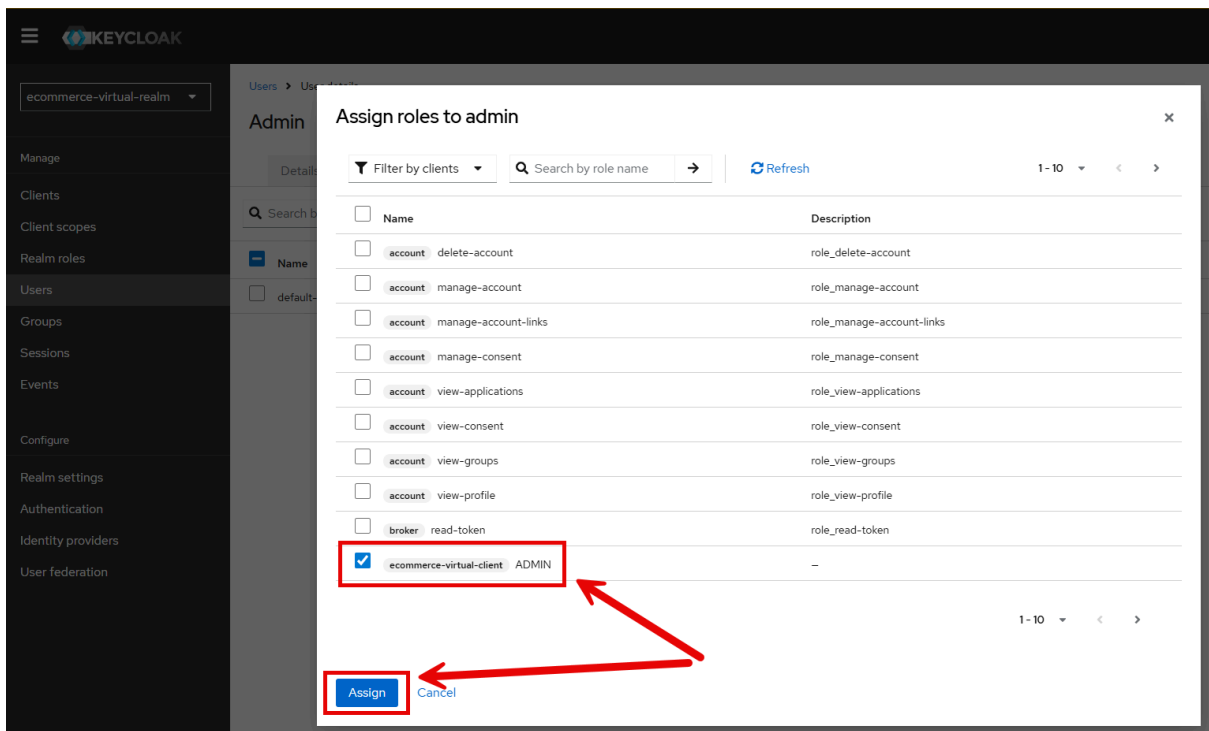
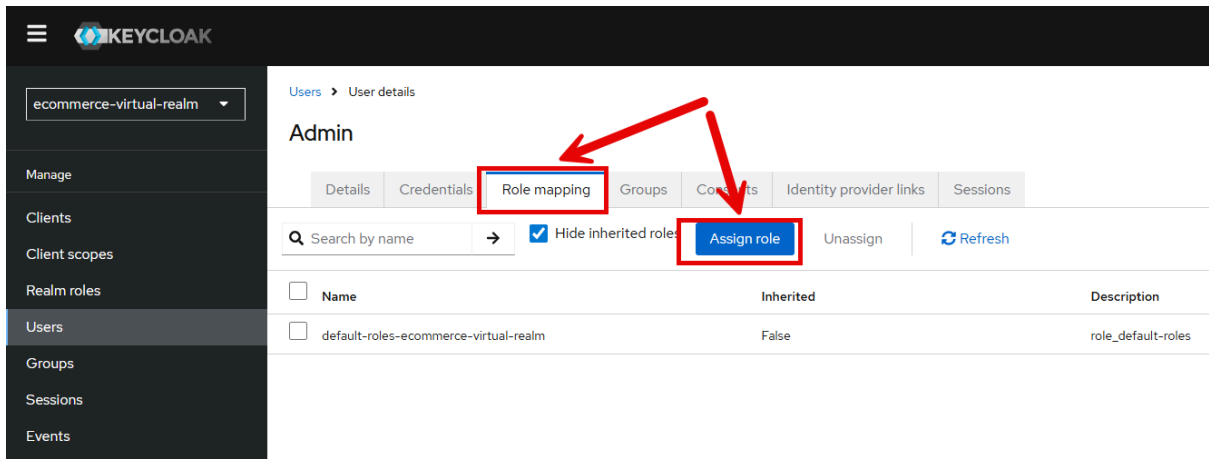
No credentials

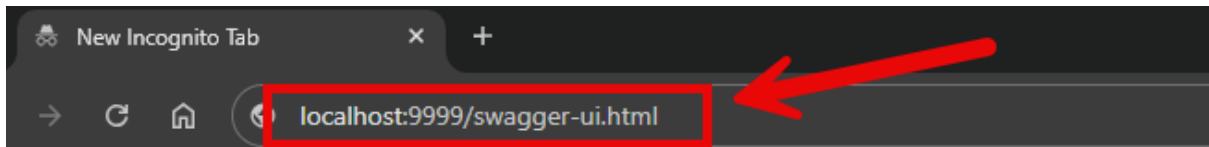
This user does not have any credentials. You can set password for this user.

Set password?

Are you sure you want to set the password for the user admin?

Save password Cancel





ECOMMERCE-VIRTUAL-REALM

Sign in to your account

Username or email

admin

Password

admin

Sign In

New user? [Register](#)

Product Service API v0.0.1 OAS 3.1

/aggregate/product-service/v3/api-docs

This is the REST API for Product Service

Apache 2.0

You can refer to the [Product Service Documentation](#)

Servers

http://localhost:8081 - Generated server url

Products REST API for Products

GET /api/v1/product Read Products

PUT /api/v1/product Update existing Product

POST /api/v1/product Create Product

Create new Product.

Parameters

No parameters

Try it out

Request body required

application/json

Example Value Schema

```
{
  "id": "1",
  "name": "Book 1",
  "description": "Description of Book 1.",
  "imageUrl": "https://upload.wikimedia.org/wikipedia/commons/8/8e/Alice%27s_Adventures_in_Wonderland_-_Carroll%2C_Robinson_-_S005_-_Alice%27s_Adventures_in_Wonderland.png",
  "locationUrl": "https://www.adobe.com/be_en/active-use/pdf/Alice_in_Wonderland.pdf",
  "price": 1.1
}
```

```
{
  "id": "1",
  "name": "Book 1",
  "description": "Description of Book 1.",
  "imageUrl": "https://upload.wikimedia.org/wikipedia/commons/8/8e/Alice%27s_Adventures_in_Wonderland_-_Carroll%2C_Robinson_-_S005_-_Alice%27s_Adventures_in_Wonderland.png",
  "locationUrl": "https://www.adobe.com/be_en/active-use/pdf/Alice_in_Wonderland.pdf",
  "price": 1.1
}
```

Execute

Clear

Responses

Curl

```
curl -X 'POST' \
  'http://localhost:8081/api/v1/product' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "id": "1",
    "name": "Book 1",
    "description": "Description of Book 1.",
    "imageUrl": "https://upload.wikimedia.org/wikipedia/commons/8/8e/Alice%27s_Adventures_in_Wonderland_-_Carroll%2C_Robinson_-_S005_-_Alice%27s_Adventures_in_Wonderland.png",
    "locationUrl": "https://www.adobe.com/be_en/active-use/pdf/Alice_in_Wonderland.pdf",
    "price": 1.1
  }'
```

Request URL

http://localhost:8081/api/v1/product

Server response

Code

Details

200

Response body

```
{
  "id": "1",
  "name": "Book 1",
  "description": "Description of Book 1.",
  "imageUrl": "https://upload.wikimedia.org/wikipedia/commons/8/8e/Alice%27s_Adventures_in_Wonderland_-_Carroll%2C_Robinson_-_S005_-_Alice%27s_Adventures_in_Wonderland.png",
  "locationUrl": "https://www.adobe.com/be_en/active-use/pdf/Alice_in_Wonderland.pdf",
  "price": 1.1
}
```

Download

 Swagger
powered by SMARTBEAR

Select a definition **Order Service**

Order Service API v0.0.1 OAS 3.1

/aggregate/order-service/v3/api-docs

This is the REST API for Order Service

Apache 2.0

[You can refer to the Order Service Documentation](#)

Servers

Orders

REST API for Orders

GET /api/v1/order Read Orders

PUT /api/v1/order Update Order

POST /api/v1/order Create Order

Create new Order. Element 'id' is created automatically. Element 'products' is mandatory. Element 'email' is mandatory only in Swagger UI.

Parameters

No parameters

Request body required

application/json

Example Value | Schema

```
{
  "id": 1,
  "email": "test@test.com",
  "products": "1,2,3",
  "orderStatus": "ORDER_INIT",
  "paymentlink": "http://payment.example",
  "token": "0123456789"
}
```

```
{
  "email": "user@example.com",
  "products": "1"
}
```

Execute

Clear

Responses

Curl

```
curl -X 'POST' \
  'http://localhost:8082/api/v1/order' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "email": "user@example.com",
    "products": "1"
  }'
```

Request URL

```
http://localhost:8082/api/v1/order
```

Server response

Code

Details

200

Response body

```
{
  "id": 1,
  "email": "user@example.com",
  "products": "1",
  "creationDate": "2026-02-25T19:09:02.7204086",
  "modificationDate": "2026-02-25T19:09:02.7204086",
  "orderStatus": "ORDER_INIT",
  "paymentlink": null,
  "token": null
}
```

 Download

You can refer to the [Payment Service Documentation](#)

http://localhost:8083 - Generated server url

```
content-type: application/json
```

mailtrap.io/box/545561-xxxxxx/5356385020

Sandboxes My Sandbox

Search...

Payment for the order number 1
to: <user@example.com> minutes ago

Payment for the order number 1
From: <ecommerce-virtual@example.com>
To: <user@example.com>
Show Headers

Looks good? Send to real users
Go live with Email API/SMTP - 4,000 production emails included

HTML HTML Source Text Raw Spam Analysis HTML Check 8 Tech Info

This is payment for the order number 1. This order consists of following items:

Item	Price	Currency
Book 1	1.1	USD
Total	1,1	USD

Payment

Book 1

\$1.10

Pay with  link

OR

Contact information

Email

user@example.com

Payment method

 Card

Card information

4242 4242 4242 4242

VISA

12 / 34

567

123

Cardholder name

Krzysztof Wisniewski

Country or region

Poland

☐ Save my information for faster checkout

Pay securely at ecommerce-virtual-sandbox and everywhere  link is accepted.

Pay

Powered by stripe | Terms Privacy

← → ↻ 🏠 ⓘ localhost:8080/api/v1/delivery?orderId=1&token=5141259712

Pretty-print ☐

```
{"message": "Delivery successful for order 1"}
```


mailtrap.io/inboxes/...

Sandboxes My Sandbox

Search...

Delivery for the order number 1
to: <user@example.com> a minute ago

From: <ecommerce-virtual@example.com>
To: <user@example.com>

Show Headers

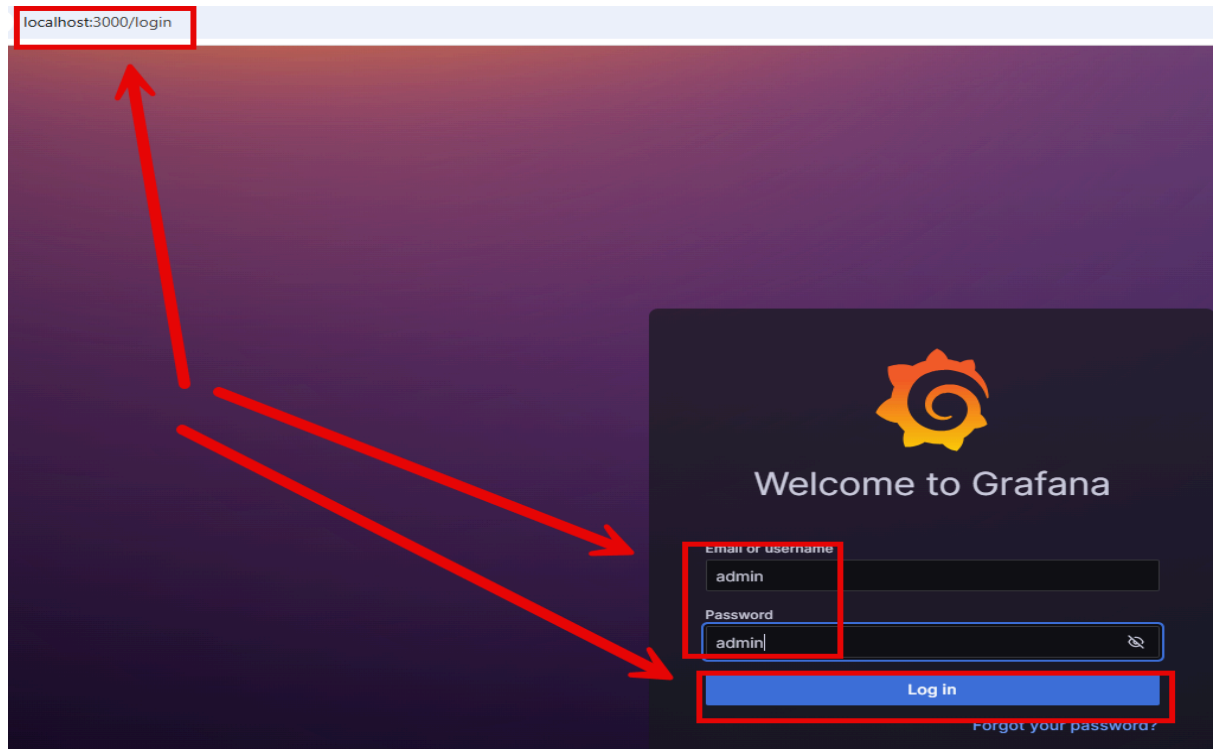
Looks good? Send it to real users
Go live with Email API/SMTP - 4,000 production emails included

HTML HTML Source Text Raw Spam Analysis HTML Check 8 Tech Info

This is delivery for the order number 1. This order consists of following items:

Item	Download
Book 1	Download

EXAMPLE GRAFANA





Update your password



Continuing to use the default password exposes you to security risks.

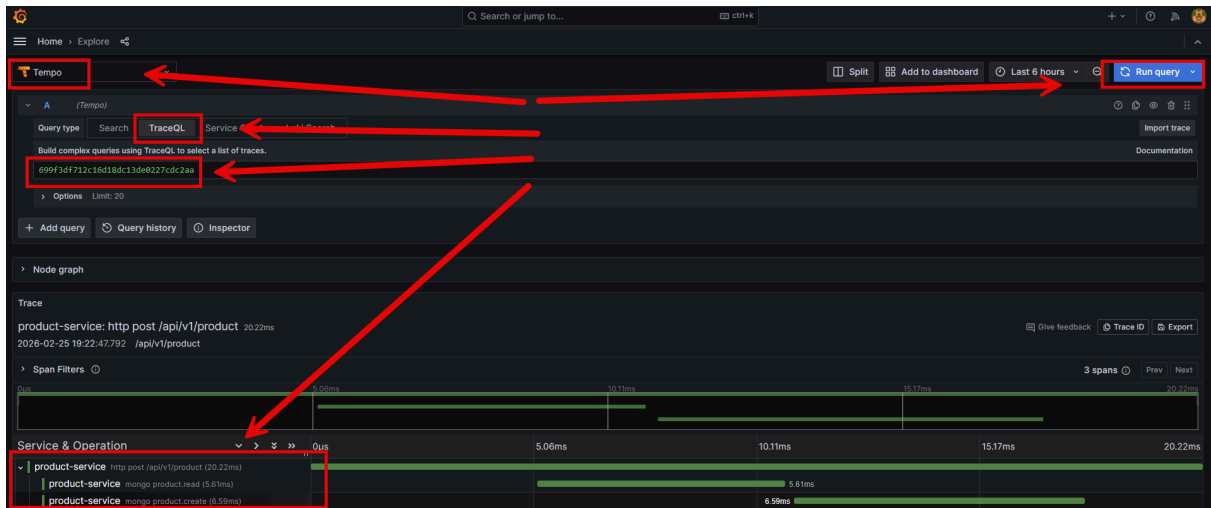
New password

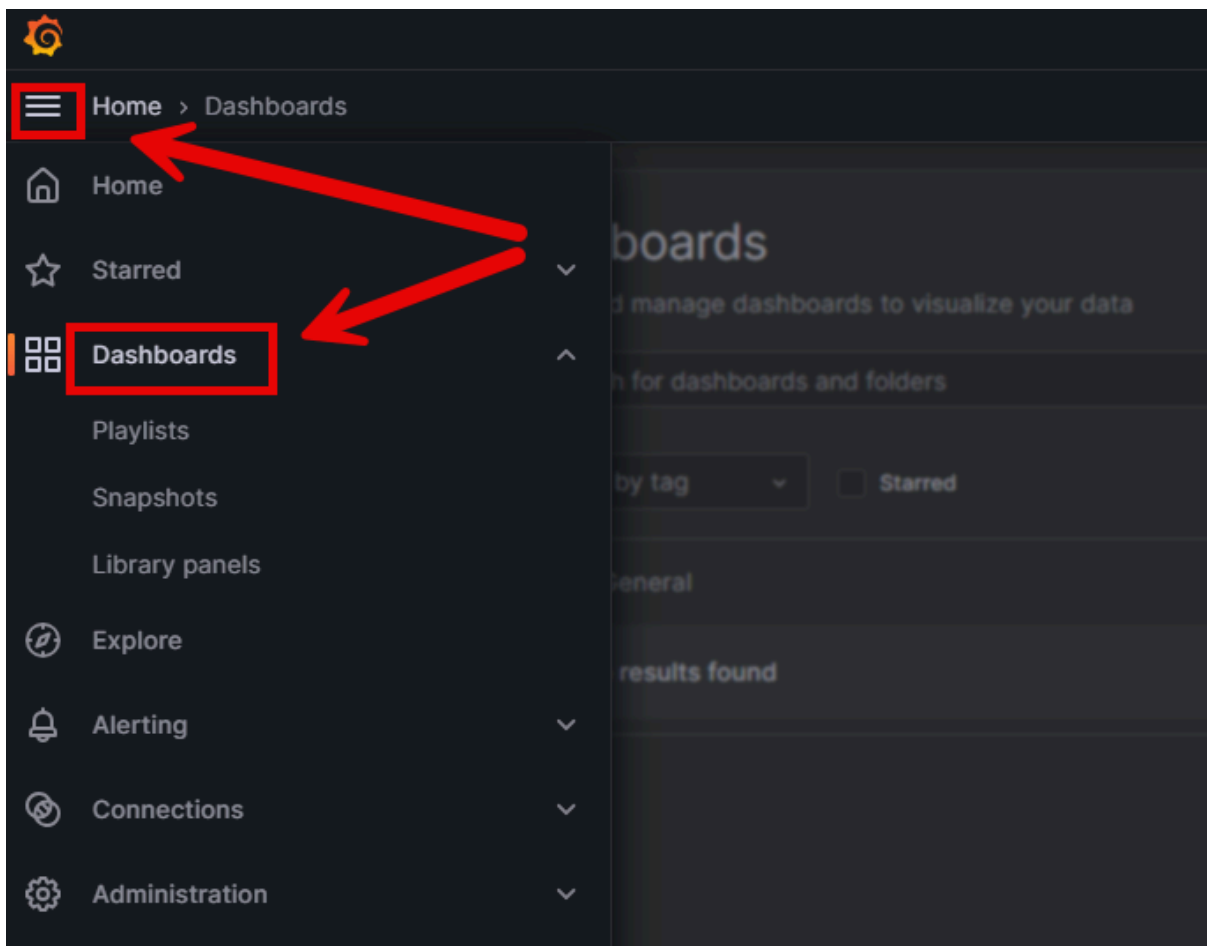
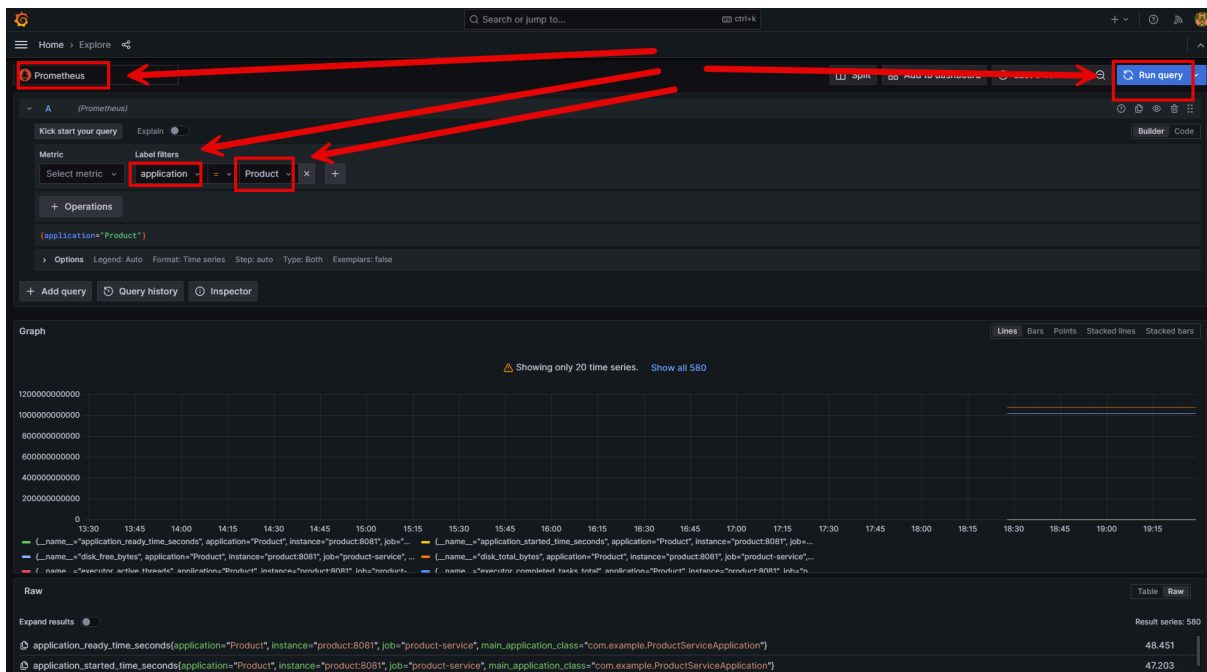
Confirm new password

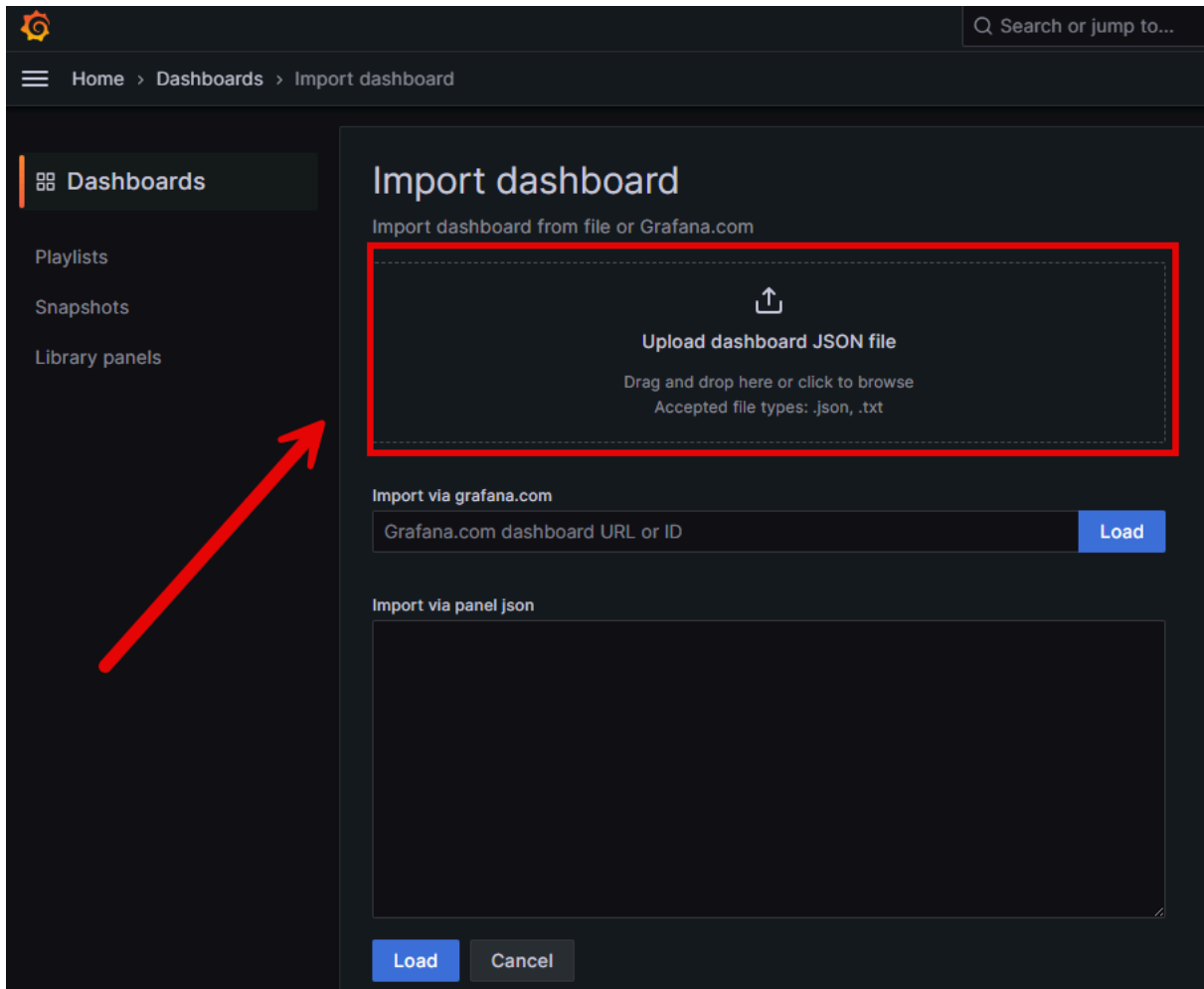
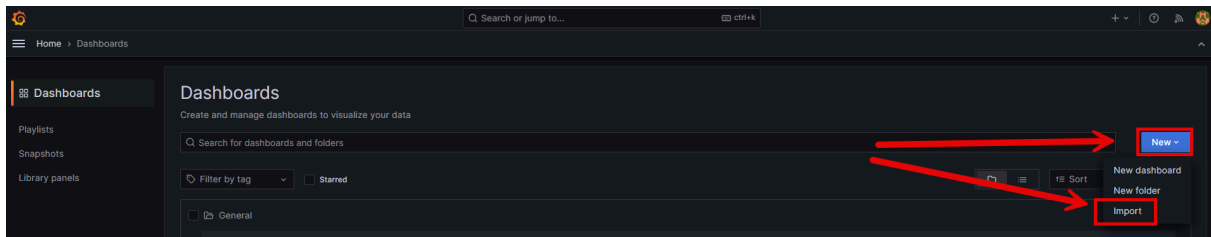
Submit

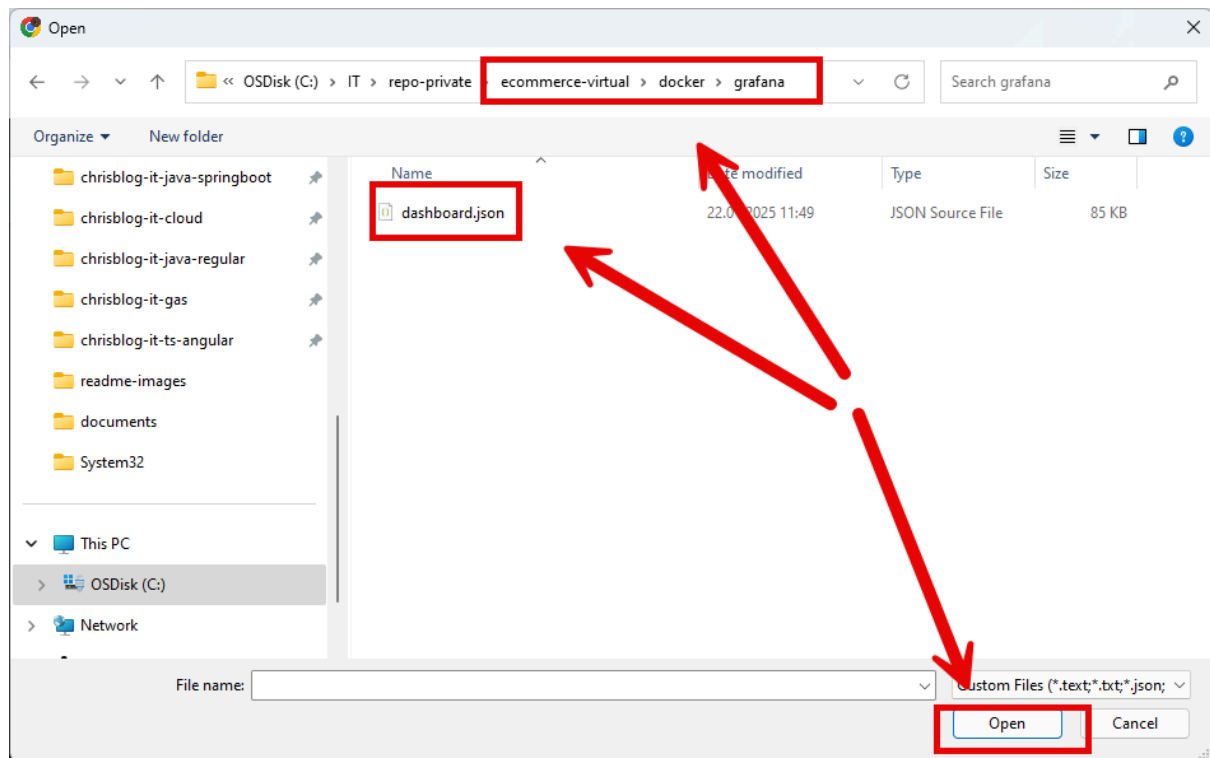
Skip











 Search or jump to...

Home > Dashboards > Import dashboard

Dashboards

Playlists

Snapshots

Library panels

Import dashboard

Import dashboard from file or Grafana.com

Options

Name

Spring Boot Statistics

Folder

General

Unique identifier (UID)

The unique identifier (UID) of a dashboard can be used to uniquely identify a dashboard between multiple Grafana installs. The UID allows having consistent URLs for accessing dashboards so changing the title of a dashboard will not break any bookmarked links to that dashboard.

sOae4vCnk

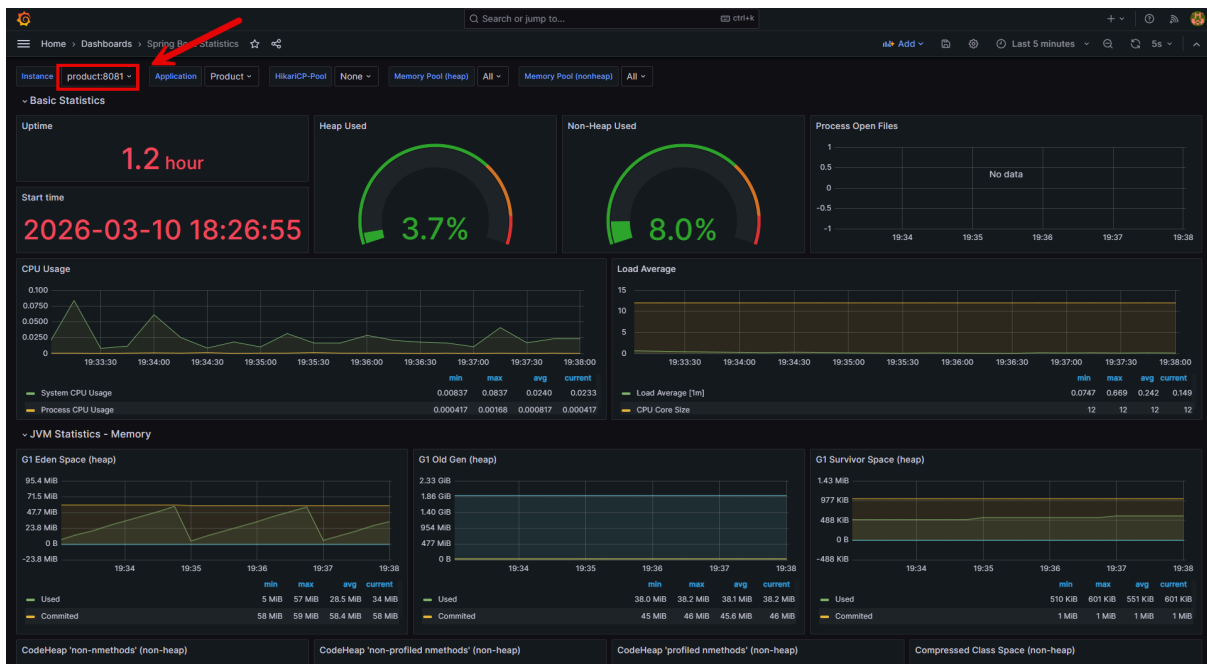
Change uid

Prometheus

Prometheus

Import

Cancel



EXAMPLE KAFKA-UI

localhost:9000/auth

Please sign in

admin

.....

Sign in

UI for Apache Kafka

83b5a60 4/10/2024, 11:41:25

Dashboard

local

Brokers

Topics

Consumers

Topics

Search by Topic Name

Show Internal Topics

+ Add a Topic

☐	Topic Name	Partitions	Out of sync replicas	Replication Factor	Number of messages	Size	
☐	_consumer_offsets	50	0	1	25	5 KB	
☐	delivery-topic	1	0	1	5	718 Bytes	
☐	payment-topic	1	0	1	4	566 Bytes	
☐	transaction-topic	1	0	1	0	0 Bytes	

UI for Apache Kafka

83b5a60 4/10/2024, 11:41:25

Dashboard

local

Brokers

Topics

Consumers

Topics

payment-topic

Overview

Messages

Consumers

Settings

Statistics

Seek Type

Offset

Offset

Partitions are selected

String

String

Clear all

Submit

Oldest First

Search

+ Add Filters

DONE

7 ms

282 Bytes

4 messages consumed

	Offset	Partition	Timestamp	Key Preview	Value Preview
+	0	0	2/25/2026, 18:58:59		..warmup..
+	1	0	2/25/2026, 19:11:21		1
+	2	0	2/26/2026, 11:39:45		..warmup..
+	3	0	2/26/2026, 11:42:23		2

CONFIGURATION OF EXTERNAL SERVICES

Before running this project, you must configure the following external services:

- **Stripe** – External service for payments.
Example configuration:
<https://github.com/wisniewskikr/chrisblog-it-java-springboot/tree/main/others/payment/springboot3-payment-stripe-api>
- **Mailtrap** – External service for e-mails.
Example configuration:
<https://github.com/wisniewskikr/chrisblog-it-java-springboot/tree/main/others/emails/springboot3-emails-text>

CONFIGURATION OF ENVIRONMENT VARIABLES

Before running this project, you must configure the following environment variables:

- **MONGO_USERNAME** – Username of the NoSQL database for products in the e-commerce shop.
- **MONGO_PASSWORD** – Password of the NoSQL database for products in the e-commerce shop.
- **POSTGRES_USERNAME** – Username of the SQL database for orders in the e-commerce shop.
- **POSTGRES_PASSWORD** – Password of the SQL database for orders in the e-commerce shop.
- **STRIPE_SECRET** – Secret key for the external payment service (provided by Stripe).
- **MAILTRAP_USERNAME** – Username for the external e-mail service (provided by Mailtrap).
- **MAILTRAP_PASSWORD** – Password for the external e-mail service (provided by Mailtrap).
- **GRAFANA_USERNAME** – Username for the UI used for distributed logging, tracking, and performance monitoring.
- **GRAFANA_PASSWORD** – Password for the UI used for distributed logging, tracking, and performance monitoring.
- **KAFKAUI_USERNAME** – Username for the UI used for asynchronous communication.
- **KAFKAUI_PASSWORD** – Password for the UI used for asynchronous communication.
- **KEYCLOAK_USERNAME** – Username for the security tool used in the e-commerce shop.
- **KEYCLOAK_PASSWORD** – Password for the security tool used in the e-commerce shop.